L02 – Distribution Network Power Flow

Foundations Lecture Series

Course Instructor

Microgrid 3-phase Security AI Project

Foundations Lecture 2

Keywords: radial topology, R/X ratio, backward/forward sweep, LinDistFlow, voltage rise

Lecture roadmap

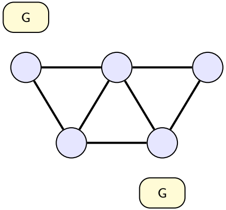
1. Why transmission and distribution need different treatments
2. Radial topology and the Backward/Forward Sweep (BFS) algorithm
3. Worked BFS example on a 4-bus feeder
4. LinDistFlow: when a linear voltage drop is enough
5. DER on distribution feeders: voltage rise and reverse flow
6. Microgrids and the bridge to L04 (unbalanced PF)

Prerequisite

L01 – you should be comfortable with Y_{bus}, Newton-Raphson, per-unit, and `pp.runpp`.

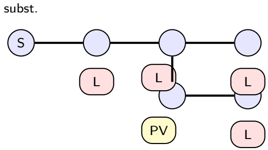
Two very different worlds

Transmission network



Meshed. Multiple generators. 110–765 kV.

Distribution feeder



Radial. One source. 0.4–35 kV. Many small loads + DER.

Why we cannot just reuse L01 directly

The math from L01 still applies in principle, but every *practical* assumption (balanced phases, low R/X, mesh topology, stable convergence) is weaker or wrong on distribution feeders.

Side-by-side comparison

	Transmission	Distribution
Voltage level	110 kV – 765 kV	0.4 kV – 35 kV
Topology	Meshed (loops)	Radial (tree)
Line R/X ratio	0.1 – 0.3	1 – 3 (sometimes higher)
Loads	Aggregated, balanced	Many small, often single-phase
Generators	Large central, three-phase	PV / BESS / EV, often single-phase
Phase symmetry	Usually fine	Rarely true
Direction of flow	One direction historically	Bi-directional with DER
Typical solver	Newton-Raphson (NR)	BFS, BFSW, sometimes NR

Reading this table

Every row is a reason this lecture exists. We will see each effect concretely on a small example before the lecture ends.

The R/X problem – with numbers

For a single radial line carrying load $P + jQ$ to the next bus, the voltage drop is approximately

$$\Delta V = V_i - V_j \;\approx\; \frac{R\,P + X\,Q}{V} \quad \text{(small-angle, low-loss approximation)}$$

Take a load of $P = 1.0$ pu, $Q = 0.3$ pu at $V \approx 1$ pu:

	R (pu)	X (pu)	ΔV (pu)
Transmission line	0.01	0.10	$0.01 + 0.03 = \mathbf{0.04}$
LV distribution feeder	0.10	0.05	$0.10 + 0.015 = \mathbf{0.115}$

The consequence

On the LV feeder ΔV is *three times larger and* dominated by the active-power term. The classical "P moves θ, Q moves $|V|$" decoupling from textbook transmission analysis no longer holds.

Why this matters for solvers

Newton-Raphson assumes...

- Reasonable initial guess (flat start works)
- Well-conditioned Jacobian
- Modest voltage drops
- Mostly inductive lines (low R/X)

On long, heavily-loaded LV feeders, the Jacobian can become ill-conditioned and NR may not converge from a flat start.

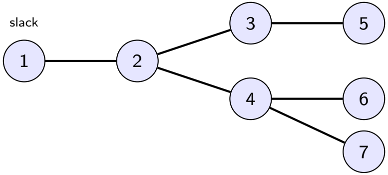
Specialized distribution methods exploit...

- Radial topology (a tree, not a graph)
- Single path from each bus to the source
- Easy KCL summation along that path
- No need to factor a large Jacobian

The most common is BFS

Backward/Forward Sweep: walk the tree twice per iteration. Simple. Robust on long radial feeders.

What "radial" means



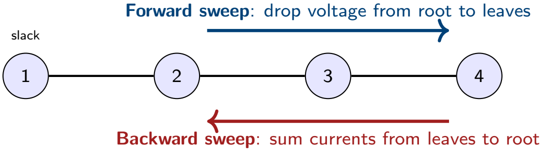
Properties of a radial network

- Exactly one path from any bus back to the slack
- No loops
- Each non-slack bus has exactly one "parent"
- Looks like a tree rooted at the substation

Why this is exploitable

- Currents downstream of bus k all flow through one branch
- KCL at each bus is a simple sum
- KVL along each branch is one equation
- No matrix inversion needed

Backward/Forward Sweep: the intuition



Two-line summary

Backward: given a voltage estimate, compute the load current at each bus and *sum* currents flowing toward the substation.
Forward: given the branch currents, walk from the slack outward and *subtract* the voltage drop on each branch.

Repeat backward + forward until the voltage estimate stops changing.

BFS algorithm in pseudocode

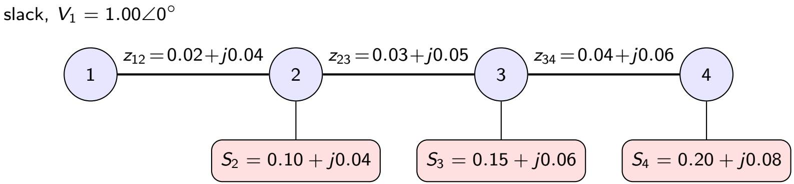
Inputs: radial topology, line impedances z_{ij}, complex loads S_i, slack voltage V_1.
Initialize: $V_i \leftarrow V_1$ for all i (flat start).

```
while not converged:
    # ---- Backward sweep ----
    # Walk from leaves to root, in reverse topological order
    for bus i in reverse_order:
        I_load_i = conj(S_i / V_i)
        I_branch[parent(i) -> i] = I_load_i + sum(I_branch[i -> child])

    # ---- Forward sweep ----
    # Walk from root outward, in topological order
    for bus j in forward_order (excluding slack):
        i = parent(j)
        V_j = V_i - z_ij * I_branch[i -> j]

    converged = max(|V_new - V_old|) < tol
```

Why this works

Radial structure guarantees `I_branch[i` $\rightarrow$ `j]` is unique: only one path connects each bus to the slack.

4-bus radial feeder: setup



All values are in per-unit on a common base

Total load: $P = 0.45$ pu, $Q = 0.18$ pu. Lines are short and resistive (typical LV feeder). Flat start: $V_2 = V_3 = V_4 = 1.0$.

Iteration 1 – Backward sweep

With $V_i = 1.0$ everywhere, the load currents are $I_i^{\text{load}} = (S_i/V_i)^*$:

Bus	S_i (pu)	I_i^{load} (pu)
2	$0.10 + j0.04$	$0.10 - j0.04$
3	$0.15 + j0.06$	$0.15 - j0.06$
4	$0.20 + j0.08$	$0.20 - j0.08$

Branch currents accumulate from the leaf back to the slack:

$$I_{3\to4} = I_4^{\text{load}} = 0.20 - j0.08$$
$$I_{2\to3} = I_3^{\text{load}} + I_{3\to4} = 0.35 - j0.14$$
$$I_{1\to2} = I_2^{\text{load}} + I_{2\to3} = 0.45 - j0.18$$

Sanity check

$I_{1\to2}$ matches total downstream load $0.45 - j0.18$. At flat start, branch currents are just summed load currents.

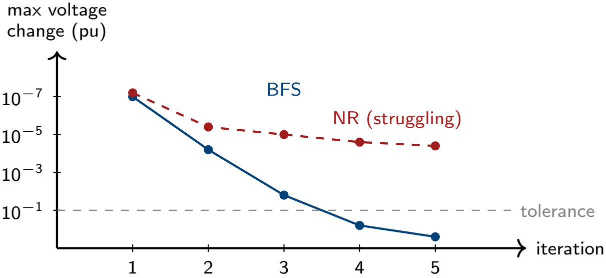
Iteration 1 – Forward sweep

Walk from the slack outward, dropping voltage by $z_{ij} \cdot I_{i\to j}$:

$$
\begin{aligned}
V_2 &= V_1 - z_{12} \cdot I_{1\to 2} = 1.000 - (0.02 + j0.04)(0.45 - j0.18) \\
&= 1.000 - (0.0162 + j0.0144) = 0.984 - j0.014 \\
V_3 &= V_2 - z_{23} \cdot I_{2\to 3} = 0.984 - j0.014 - (0.03 + j0.05)(0.35 - j0.14) \\
&= 0.984 - j0.014 - (0.0175 + j0.0133) = 0.966 - j0.028 \\
V_4 &= V_3 - z_{34} \cdot I_{3\to 4} = 0.966 - j0.028 - (0.04 + j0.06)(0.20 - j0.08) \\
&= 0.966 - j0.028 - (0.0128 + j0.0088) = 0.954 - j0.037
\end{aligned}
$$

Bus	V_i (complex pu)	$\lvert V_i \rvert$ (pu)
2	$0.984 - j0.014$	0.984
3	$0.966 - j0.028$	0.967
4	$0.954 - j0.037$	0.954

Convergence in a few sweeps



BFS typically

- 3–6 iterations to 10^{-8}
- Cheap per iteration (no Jacobian)
- Reliable on long radial feeders

NR on the same problem

- May not converge from flat start
- Each iteration more expensive
- Better on meshed transmission

Illustrative; exact convergence rate depends on R/X, loading, and topology.

When BFS beats NR (and when it does not)

Scenario	BFS	NR
Long radial LV feeder, heavy load	✓	△
High R/X (e.g. 1–3)	✓	△
Lightly-meshed distribution	✓ *	✓
Heavily-meshed transmission	△	✓
PV-rich feeder with reverse flow	✓ *	✓
Need exact derivatives (sensitivity studies)	△	✓

✓ * requires BFSW (Backward/Forward Sweep with Walks) or other generalizations.

In pandapower

```
pp.runpp(net, algorithm="bfsw") # backward/forward sweep with walks
pp.runpp(net, algorithm="nr") # default Newton-Raphson
```

Try both on the same network. On a long LV feeder you will often see one converge while the other does not.

The exact branch equation

For a radial branch from bus i to bus j carrying complex power $P_{ij} + jQ_{ij}$ at the sending end:

$$|V_i|^2 - |V_j|^2 \;=\; 2\,(R_{ij}\,P_{ij} + X_{ij}\,Q_{ij}) \;-\; |Z_{ij}|^2 \cdot |I_{ij}|^2$$

Two terms:

- $2(RP + XQ)$ – the dominant "voltage drop" term, linear in P, Q.
- $|Z|^2 \cdot |I|^2$ – the "loss" term, quadratic in current. Small when V is near 1 and losses are a few percent.

LinDistFlow (Baran & Wu, 1989)

Drop the loss term. Then linearize $|V_i|^2 - |V_j|^2 \approx 2(|V_i| - |V_j|)$ around $|V| \approx 1$:

$$\boxed{|V_i| - |V_j| \;\approx\; R_{ij}\,P_{ij} + X_{ij}\,Q_{ij}}$$

A clean linear formula. Used everywhere in distribution planning and optimization.

LinDistFlow on the worked example

Apply LinDistFlow to the same 4-bus feeder, using P_{ij}, Q_{ij} = total downstream load on each branch:

Branch	P_{ij} (pu)	Q_{ij} (pu)	$\Delta V = RP + XQ$	$\lvert V_j \rvert^{\text{LinDist}}$
$1 \to 2$	0.45	0.18	$0.02(0.45) + 0.04(0.18) = 0.016$	0.984
$2 \to 3$	0.35	0.14	$0.03(0.35) + 0.05(0.14) = 0.017$	0.967
$3 \to 4$	0.20	0.08	$0.04(0.20) + 0.06(0.08) = 0.013$	0.954

Bus	$\lvert V \rvert$ from BFS	$\lvert V \rvert$ from LinDistFlow
2	0.984	0.984
3	0.967	0.967
4	0.954	0.954

Why they agree so well

On this feeder, voltages are close to 1.0 pu and losses are a fraction of a percent. The neglected $|Z|^2|I|^2$ term is in the 4th decimal place.

When LinDistFlow is accurate, and when it lies

Accurate within $\sim 0.5\%$ when:

- All $|V_i| \in [0.95, 1.05]$
- Losses $<$ 2–3% of total flow
- No large reverse flow
- No dominant $|I|^2$ heating

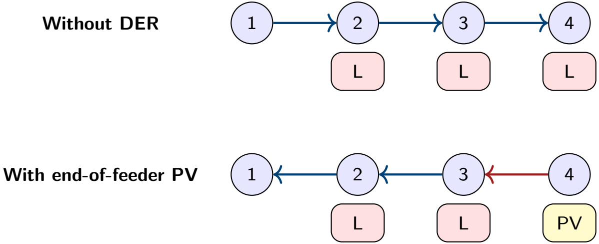
Becomes optimistic when:

- Heavy PV generation pushing voltages above 1.05
- Long feeder with $|V|$ near 0.92–0.93
- High line current $\rightarrow$ $|I|^2 R$ losses matter
- Capacitor banks switching $\rightarrow$ Q changes abruptly

Where you will see it

LinDistFlow underlies many distribution OPF formulations, DER siting/sizing problems, and Volt/VAR optimization. We will use BFS in this course because we want absolute voltage values, but the linear form is what makes large planning problems tractable.

What DER changes



- Without DER: power flows substation $\rightarrow$ loads. Voltage decreases away from substation.
- With heavy DER at the end of the feeder: power can flow *toward* the substation. Voltage *rises* near the DER.

Voltage rise – a numerical example

A 2-bus toy feeder, LV characteristics: $z = 0.05 + j0.05$ pu, $V_1 = 1.00$.

Case	Net injection at bus 2	$\Delta V = (RP + XQ)/V$	$\lvert V_2 \rvert$
(a) 0.5 pu load only	-0.50 pu	$+0.025$	**0.975** pu
(b) 0.5 pu PV, 0.5 pu load (offset)	0.00 pu	0.000	**1.000** pu
(c) 1.5 pu PV, 0.5 pu load (export 1.0)	$+1.00$ pu	-0.050	**1.050** pu
(d) 2.0 pu PV, 0.5 pu load (export 1.5)	$+1.50$ pu	-0.075	**1.075** pu

The pattern

Case (a) is the classical low-voltage problem: load $\rightarrow$ voltage drops.
Case (d) is the modern high-voltage problem: PV $\rightarrow$ voltage rises past 1.05 pu.
Same feeder, opposite violations. A simple "low-voltage check" is no longer enough.

Reverse power flow and what it breaks

When DER injection at a bus exceeds local load, net power flows *toward* the substation. Several consequences:

- Voltage rises along the upstream path.
- Some protection relays assume one-directional flow.
- Transformer tap controllers may move the wrong way.
- Loss calculations need to be reconsidered.

Magnitude in practice

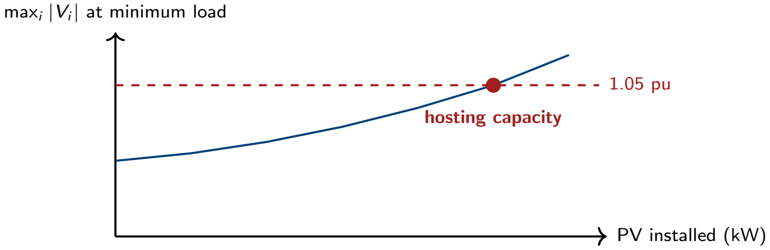
At 30–50% PV penetration on a typical LV feeder, the substation can already see negative net flow at midday on clear weekends.

Connection to this course

Reverse flow + voltage rise = a violation mode the N-1 screening model must learn. Week 4 of the project labels both kinds.

Hosting capacity in one slide

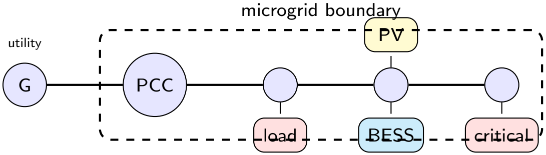
Question: how much DER can be added to a feeder before *any* voltage or thermal limit is violated?



Definitions

- **Hosting capacity** = the PV size at which the first constraint becomes binding under the worst-case operating condition (typically: maximum PV, minimum load).
- It is determined by either voltage rise or thermal overload, whichever comes first.
- Computing it requires the same N-1-style PF scans we will use in Weeks 3–4 of the project.

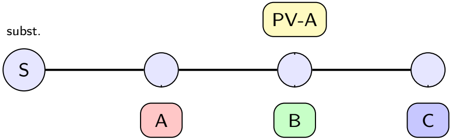
Microgrid = distribution + control boundary



What “microgrid” adds on top of a distribution feeder

- Defined electrical boundary at the Point of Common Coupling (PCC).
- Internal generation that can support load when the utility is disconnected.
- Supervisory control (energy management system).
- Often: critical loads with higher reliability requirements.

Why distribution is rarely truly balanced



Sources of phase asymmetry on LV / microgrid feeders:

- Single-phase loads (most residential consumers connect to one phase only).
- Single-phase rooftop PV (the inverter is connected to the homeowner's phase).
- Single-phase EV chargers, water heaters, ovens.
- Untransposed lines – transposition is rarely worth the cost at LV.
- Asymmetric three-phase loads (one phase under heavy use, others idle).

Implication

Phase A could drop to 0.92 pu while phase B sits at 1.01 pu and phase C at 1.03 pu. *Average* voltage looks fine. Balanced power flow happily reports "no violation". Phase A is in violation.

Teaser of L04

What L04 will add to today's machinery:

- Symmetrical components: positive, negative, zero.
- Sequence-domain formulation of the PF.
- Voltage Unbalance Factor (VUF).
- `pandapower.runpp_3ph` as the practical solver.
- Five violation types (incl. VUF).

What stays from L02

- Radial topology still simplifies things.
- BFS generalizes to three-phase.
- Voltage rise behavior still applies per-phase.
- Hosting capacity becomes per-phase.

Bridge

L02 said: "transmission tools do not transfer to distribution." L04 will say: "single-phase distribution tools do not transfer to phase-aware distribution." Same kind of failure of an unstated assumption.

Homework

Required

1. Reproduce the 4-bus radial example in pandapower. Use `pp.create_line_from_parameters` with R, X consistent with the per-unit values in the slides.
2. Run both `pp.runpp(net, algorithm="bfsw")` and `pp.runpp(net, algorithm="nr")`. Confirm `net.converged` for both and verify $|V_i|$ matches the worked example to 3 decimal places.
3. Multiply all line impedances by 5. Re-run. Which bus moves the most? Why?
4. Replace the load at bus 4 with a 1.0 pu PV `sgen`. Find the smallest PV injection that pushes $|V_4|$ above 1.05.

Optional

1. Implement BFS yourself in ∼30 lines of `numpy` and compare to pandapower's `bfsw`.
2. Plot the voltage profile $|V_i|$ vs bus index for load levels 0%, 50%, 100%, 150% of the slide values.
3. Find the hosting capacity (max PV at bus 4 before any violation) under the worst-case minimum load.

Exit checklist

1. ✓ I can explain in one sentence why transmission solvers struggle on distribution feeders.
2. ✓ I can sketch the side-by-side topology of a transmission vs distribution network.
3. ✓ I can write the simplified voltage drop $\Delta V = (RP + XQ)/V$ and explain when it lies.
4. ✓ I can describe one backward sweep and one forward sweep in BFS on a small radial feeder.
5. ✓ I can choose between `algorithm="nr"` and `algorithm="bfsw"` for a given network.
6. ✓ I can describe how end-of-feeder PV causes voltage rise.
7. ✓ I can define hosting capacity in two sentences.
8. ✓ I know exactly which assumption from L02 will break in L04 (single-phase equivalent).

References

- **Kersting,** *Distribution System Modeling and Analysis* (4th ed., 2017). The canonical distribution PF textbook.
- **Baran & Wu, 1989.** *Optimal Capacitor Placement on Radial Distribution Systems*, IEEE TPD. The DistFlow / LinDistFlow paper.
- **Cespedes, 1990.** *New Method for the Analysis of Distribution Networks*, IEEE TPD. Classical BFS reference.
- **Shirmohammadi et al., 1988.** *A Compensation-Based Power Flow Method for Weakly Meshed Distribution Networks*, IEEE TPS.
- **Bollen & Hassan,** *Integration of Distributed Generation in the Power System* (Wiley, 2011). Voltage rise, hosting capacity.
- **IEEE Std 1547-2018**, *Standard for Interconnection of DER*. Voltage and frequency ride-through requirements.
- **pandapower docs:** `https://pandapower.readthedocs.io` – look for “BFSW” under power-flow algorithms.